INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR
Howrah, West Bengal, India – 711103

DEPARTMENT OF COMPUTER SCIENCE AND TECHNOLOGY



A PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS

ON

"DETECTION OF CIPHER TYPES USING ARTIFICIAL NEURAL NETWORKS"

SUBMITTED BY

Rajdeep Ghosh (510519002)
Swarnali Mondal (510510103)
Krishnendu Bera (510519004)
Arnab Sen (510519006)
Sanjana Chakravarty (510519038)
Abhiroop Mukherjee (510519109)

UNDER THE GUIDANCE OF

DR. MALAY KULE

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INDIAN INSTITUTE OF ENGINEERING SCIENCE AND TECHNOLOGY, SHIBPUR

Howrah, West Bengal, India – 711103

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Certificate

It is certified hereby that this report, titled "Detection of Cipher Types using Artificial Neural Networks", and all the attached documents herewith are authentic records of Rajdeep Ghosh (510519002), Swarnali Mondal (510510103), Krishnendu Bera (510519004), Arnab Sen (510519006), Sanjana Chakravarty (510519038), and Abhiroop Mukherjee (510519109) from the Prestigious Department of Computer Science And Technology of the Distinguished and Respected IIEST Shibpur under my guidance.

The works of these students satisfy all the requirements for which it is submitted. To the extent of my knowledge, it has not been submitted to any different institutions for the award of a degree.

Dr. Malay Kule
Asst. Professor

Dr. Asit Kumar Das
Head Of Department

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Rajdeep Ghosh (510519002)
Swarnali Mondal (510510103)
Krishnendu Bera (510519004)
Arnab Sen (510519006)
Sanjana Chakravarty (510519038)
Abhiroop Mukherjee (510519109)

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1 INTRODUCTION

Information is the power in the present day. Information and data should be protected from unauthorized access, distortion, or any unauthorized activity.

Cryptography is the study of secret writing. Information is no longer hidden due to the existence of network adversaries. Our communication channel is completely unstable; a third party might covertly eavesdrop and obtain information from the network or from the channel. Here, cryptography is crucial in preventing unauthorized access to our data. Using key(s), cryptography converts plaintext messages into unreadable ciphertext and vice versa. Encryption is the process of employing a key(s) to transform the original message into an unintelligible format, while decryption is the act of using a key(s) to convert the cipher text back to plaintext. The messages can only be encrypted or decrypted by the authorized user.

The project's objective is to examine encrypted text files to find and categorize their cipher types using artificial neural networks. The cipher algorithm classification model is based on the steps:

1. Creation of a collection of cipher texts with various cipher texts.
2. Building a Deep Neural Network (DNN) and training it with a variety of cipher texts and training parameters to enable accurate learning and cipher text classification.
3. Using the trained model to determine the algorithm that might have been used to generate the cipher text.

2 PRELIMINARIES

To ramp up for this project, we needed a basic knowledge of neural networks and deep learning and an overview of the classical encryption algorithms that we would be targeting in this project. Having specific knowledge of the encryption algorithms will be useful to particularly train the model.

The ML topics that are covered in this project are:

- Perceptron
- Artificial Neural Network (ANN)
- Recurrent Neural Network (RNN)
- Convolutional Neural Network (CNN)
- Transformers
- Attention Mechanism in Deep Learning

3 LITERATURE SURVEY

Over recent years, artificial neural networks (ANNs) have come to prominence. Applications and research areas for ANNs are numerous and diverse. Their primary objective is the quick filtering, classification, and processing of (mainly) non-linear data, such as speech recognition, picture processing, and language translation. In addition, researchers were able to "train" ANNs to play games (e.g. AlphaStar [1]) or produce artwork in the manner of well-known artists.

On the MysteryTwister C3[2] (MTC3) website in late 2019, Stamp posted a challenge titled "Cipher ID". The task is to identify the type of cipher used to encrypt each of the 500 ciphertexts in the collection, which were encrypted using 5 distinct ciphers and random keys. The individual ciphertexts were then mixed up after each cipher type had been employed exactly 100 times. All prior solvers completed the task by deciphering the ciphertexts using solvers for the 5 various cipher types, despite the author's goal of encouraging others to begin research in the fields of machine learning and cipher type detection. The participants were able to determine the type of encryption the algorithm that was employed after learning the plaintext of each cipher.

We worked on the cipher text detection problem with the help of ANN and in this process, TensorFlow and Pytorch were used. TensorFlow is a free and open-source data flow and math library developed by Google written in Python, C++, and CUDA, and was publicly released in 2015. PyTorch is an open-source machine learning (ML) framework based on the Python programming language and the Torch library which was initially released in the year 2016.

In this section, we present different papers and articles, which we studied before starting with our work. The usage of ANNs in the paper ranges between the emulation of ciphers, the detection of the cipher type, and the recovery of cryptographic keys.

1. Attention Is All You Need[3] (NIPS 2017), here a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

2. Detecting the Type of Ciphers Using Artificial Neural Networks[4] (ICHC, 2020), where a first version of an artificial neural network was developed to differentiate between five classical ciphers: simple monoalphabetic substitution, Vigenere, Playfair, Hill, and transposition.

3. In the Language Models are Unsupervised Multitask Learners[5] paper, demonstration of language models that can perform down-stream tasks in a zero-shot setting – without any parameter or architecture modification was done.

4. Language Models are Few-Shot Learners[6], broader societal impacts of this finding and of GPT-3 in general are discussed in this paper

5. Detection of Classical Cipher Types with Feature-Learning Approaches[7], where the applicability of the two feature-learning algorithms, long short-term memory (LSTM) and Transformer, have been explored to overcome the dependency on domain knowledge

6. A systematic evaluation of large language models of code[8] (ICLR 2022), where a new model PolyCoder was released, with 2.7B parameters based on the GPT-2 architecture, that was trained on 249GB of code across 12 programming languages on a single machine.

7. GPT-NeoX-20B: An Open-Source Autoregressive Language Model[9], where GPT-NeoX-20B was introduced which was a 20-billion-parameter autoregressive language model trained on the Pile.

8. Fine-tune a pretrained model[10], a tutorial where a step-by-step process to fine-tune a pretrained model with a deep learning framework is discussed.

9. Fine-Tuning Transformers for NLP[11], another tutorial on how to fine-tune two different transformer models, BERT and DistilBERT: Sentiment Analysis, and Duplicate Question Detection is covered.

10. How to generate text: using different decoding methods for language generation with Transformers [12], a blog post which gives a brief overview of different decoding strategies and shows how can they be implemented with very little effort using the popular transformers library

4 PROPOSED WORK

4.1 Dataset Generation

The dataset has been generated for five types of ciphers:

- Caesar
- Affine
- Vigenère
- Substitution
- Rail Fence

The plaintext used for encryption was a text from Shakespeare's Hamlet and Macbeth.

For Data Preprocessing, we converted text to lowercase and removed all whitespaces and punctuation.

Each of the classes has 6815 data points, resulting in a total of 34075 data points and each data point consists of the ciphertext (input) and the encryption algorithm (output).

We wrote a Python script for each of the ciphers to encrypt the data into a Pandas DataFrame. The combined DataFrame is shuffled and exported as a CSV file for the resulting dataset.

4.2 Model using CNN

The main intuition behind this method is that CNN uses its kernel as a "local receptive field", which means that a CNN kernel (small in size as we are dealing with stream cipher and not block cipher) can look at its nearby characters and will find some pattern in the ciphertext

So, we first converted every ciphertext into their respective list of tokens using a TextVectorizer, then we converted that list into a 2D text matrix as follows:

$$\begin{bmatrix} a & b & c & d \end{bmatrix} \longrightarrow \begin{bmatrix} a & b & c & d \\ a & b & c & d \\ a & b & c & d \\ a & b & c & d \end{bmatrix}$$

Figure 1: Converting text to 2D text matrix

Note that this conversion depends on the length of the string used, i.e. for a string for length n, we will generate a $n \times n$ matrix". After we generate all variable-sized matrices, we resize them to 60×60 using Bilinear Interpolation.

Now we can see that a kernel (say 3×3 with stride one) can observe its previous as well as its next character and can use that information to find patterns in the ciphertext.

We use a CNN to extract features from the matrix and then we feed those features to a Dense Network to classify between the ciphertext types.

4.3 Model using Transformer

The main intuition behind this method is to compute the relative importance of a ciphertext character with respect to the other ciphertext characters. A transformer model works on the encoder decoder architecture converting ciphertext characters to tokens and then computing the relative importance of a token to other tokens.

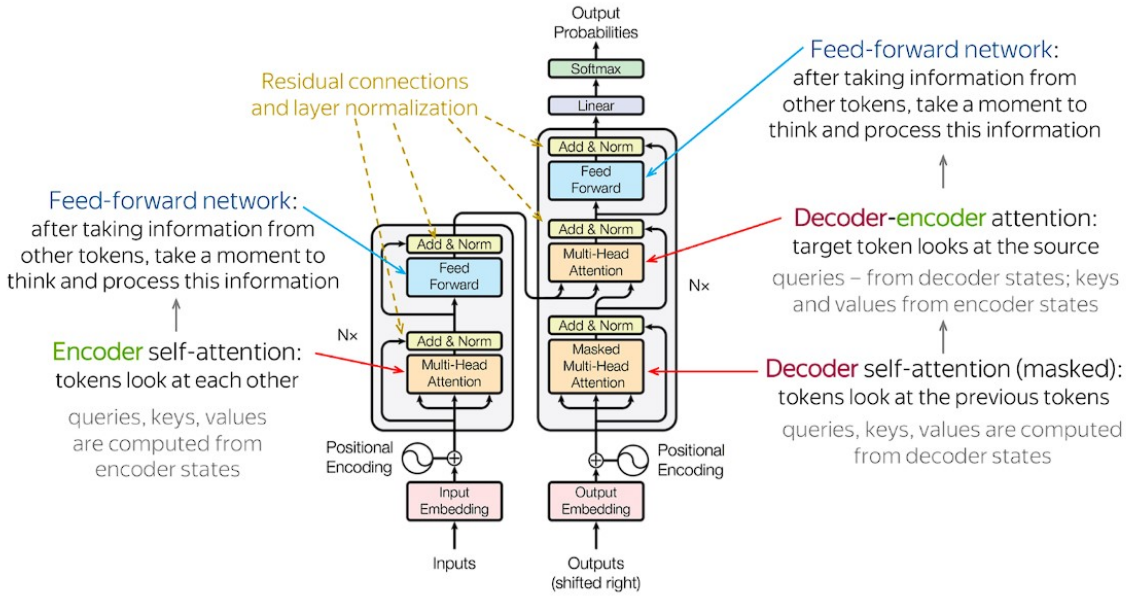


Figure 2: Transformer architecture

A transformer is an advanced sequence-to-sequence model that is used to predict the next token depending on all the previous tokens. From our dataset containing encrypted text and the algorithm used for encryption, we create a text prompt from each row for example:

Cipher text: `ymccaygimgamkyyiycuycqocyeagmigkycmyyoyy`
Algorithm: `affine`

The model learns the relative importance of each token and uses it to predict the next. For example, if the model is trained with the above prompt and then we supply the following prompt:

Cipher text: `ymccaygimgamkyyiycuycqocyeagmigkycmyyoyy`
Algorithm:

The model shall generate the output as “affine” which it learned from training.

In our use case, we have used the GPT-Neo (Generative Pre-Trained) model with 125 million parameters to achieve our objective. It consists of stacks of encoders and decoders put one after the other, of which has been pre-trained on Wikipedia Corpus as well as Common Crawl (This has over 12 Petabytes of data which is 12 years of data uploaded on the internet) datasets for performing extremely well on language-based use cases. Generative, as the word suggests, is for making our code generate text. The architecture is similar to GPT2 except that GPT Neo uses local attention in every other layer with a window size of 256 tokens. All these models are available for free in the Hugging Face transformers library which can be used for a variety of NLP tasks.

The following steps were taken for this model:

1. Each data of the dataset is converted into text prompts that is passed through a tokenizer that creates embedding for each token in our input string.
2. The input dataset has been split into the ratio 80:10:10 for training, testing, and validating purposes.
3. The training and validation set has been used to fine-tune the model's hyperparameters specific to our use-case i.e. detecting the algorithm used in the ciphertext generation.
4. We used the fine-tuned model on our test dataset to check the accuracy of our model.

For our testing purpose, we used various decoding strategies to find the one which gives the best result. Out of the strategies: Greedy Search, Beam Search, and Sampling with set temperature, we found sampling to give the most accurate results of all three. It is a method that is in between the greedy and sampling approach.

The greedy method selects the next token that has the highest probability as the output while sampling selects one of the possible next tokens randomly. When we set a temperature, the randomness is lowered such that the likelihood of high-probability words are increased and that of the low probability are decreased. We found that a **temperature set to 0.7** gave the optimum accuracy on our test data.

5 EXPERIMENTAL RESULTS

5.1 Model using CNN

Our dataset consisted of 34075 rows that were split for training, testing, and validating in the ratio 80:10:10. The following are the sizes for each split:

- Train dataset: 27260 rows
- Validation dataset: 3408 rows
- Test dataset: 3407 rows

We also added an EarlyStopping Callback of patience 10 which automatically stopped training if we do not see any validation accuracy improvement after 10 epochs.

```
Epoch 1/10000
852/852 [==============================] - 11s 10ms/step - loss: 1.1977 - accuracy: 0.4627 - val_loss: 1.0805 - val_accuracy: 0.5161
Epoch 2/10000
852/852 [==============================] - 9s 10ms/step - loss: 1.0930 - accuracy: 0.5132 - val_loss: 1.0625 - val_accuracy: 0.5174
Epoch 3/10000
852/852 [==============================] - 9s 10ms/step - loss: 1.0494 - accuracy: 0.5465 - val_loss: 0.9777 - val_accuracy: 0.5789
Epoch 4/10000
852/852 [==============================] - 9s 10ms/step - loss: 1.0158 - accuracy: 0.5716 - val_loss: 0.9707 - val_accuracy: 0.5864
Epoch 5/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.9896 - accuracy: 0.5856 - val_loss: 1.0243 - val_accuracy: 0.5668
Epoch 6/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.9701 - accuracy: 0.5977 - val_loss: 0.9334 - val_accuracy: 0.6190
Epoch 7/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.9557 - accuracy: 0.6036 - val_loss: 1.0162 - val_accuracy: 0.5668
Epoch 8/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.9384 - accuracy: 0.6118 - val_loss: 0.9276 - val_accuracy: 0.6102
Epoch 9/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.9172 - accuracy: 0.6206 - val_loss: 1.1872 - val_accuracy: 0.5215
Epoch 10/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.9054 - accuracy: 0.6286 - val_loss: 0.9491 - val_accuracy: 0.6086
Epoch 11/10000
852/852 [==============================] - 9s 10ms/step - loss: 0.8890 - accuracy: 0.6372 - val_loss: 0.9704 - val_accuracy: 0.5879
Epoch 12/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.8593 - accuracy: 0.6549 - val_loss: 1.0264 - val_accuracy: 0.5836
Epoch 13/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.8401 - accuracy: 0.6607 - val_loss: 0.9895 - val_accuracy: 0.5970
Epoch 14/10000
852/852 [==============================] - 9s 10ms/step - loss: 0.8155 - accuracy: 0.6744 - val_loss: 0.9790 - val_accuracy: 0.6005
Epoch 15/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.7853 - accuracy: 0.6884 - val_loss: 1.0477 - val_accuracy: 0.5800
Epoch 16/10000
852/852 [==============================] - 8s 10ms/step - loss: 0.7653 - accuracy: 0.6952 - val_loss: 1.0013 - val_accuracy: 0.5967
```

Figure 3: Training Log for the 5-class Classification Model

Here we can see that as Training Stopped at epoch 16, our best model was in epoch 6 which had a validation accuracy of 61.90%

After some trial and error, we found that the best model we could make by altering the network size, had 2.7 million parameters for the five-class classification and 20.4 million parameters for the four-class model

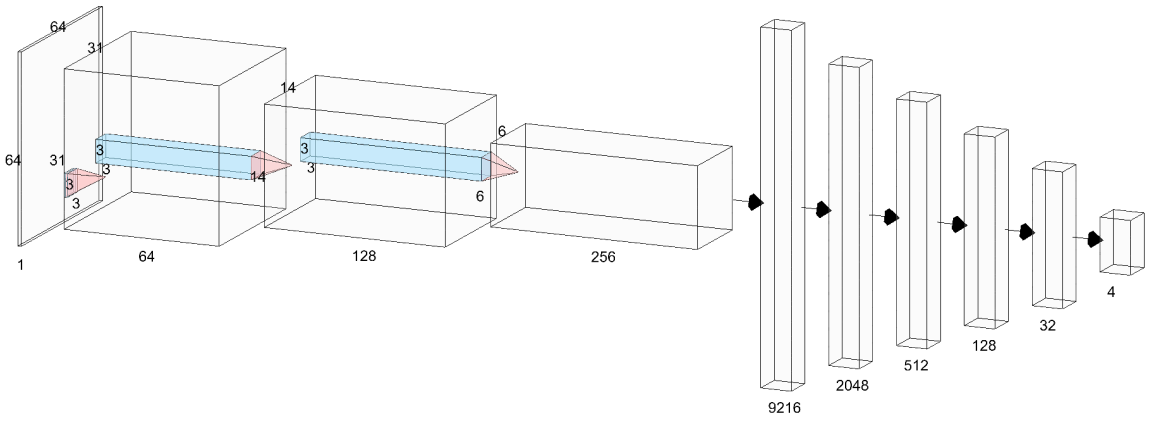


Figure 4: Model Diagram for 4 Class Classifier

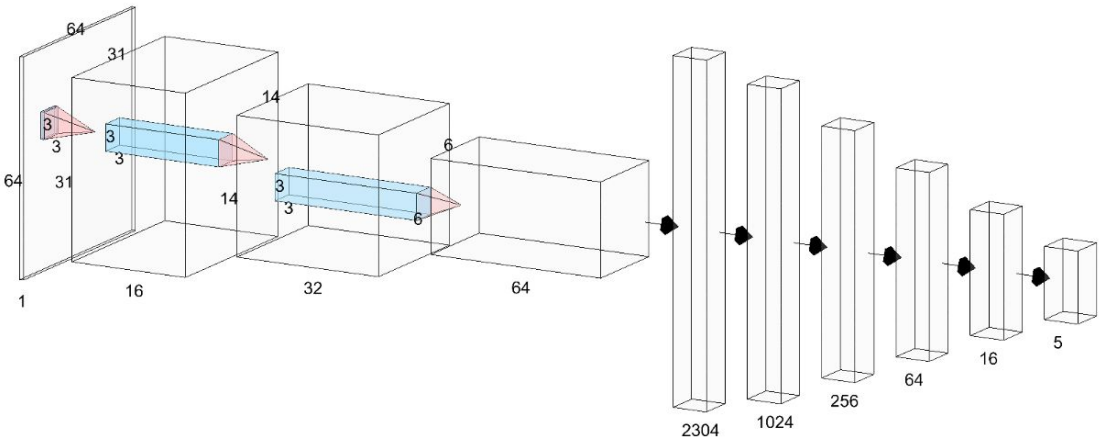


Figure 5: Model Diagram for 5 Class Classifier

We were getting the test accuracy **61.90% for four class classifier and 59.71% for the five-class classifier**

5.2 Model using Transformer

Here we used GPT-neo with 125 million parameters and fine-tuned it with our generated dataset.

Our dataset consisted of 34075 rows that were split for training, testing, and validating in the ratio 80:10:10. The following are the sizes for each split:

- Train dataset: 27260 rows
- Validation dataset: 3408 rows
- Test dataset: 3407 rows

In the beginning, each of our input prompts is converted into tokens using GPT2 tokenizer.

```
print((final_test_dataset[5]))
print(test_dataset[5])
```

```
Cyphertext: ymccaygimgamkyyiycuycqocyeagmigkycmyyoyy
Algorithm: affine

(tensor([   34,  4464,   372,  5239,    25,   331,    76,   535,   323,    70,
         9600,   321,  2584,    88,  7745,    66,  4669,    66,    80,   420,
         5948,   363,    76,   328,  2584,    66,  1820,    88,   726,    88,
          198,  2348, 42289,    25,  1527,   500,   198, 50256, 50256, 50256,
        50256, 50256, 50256, 50256, 50256, 50256, 50256, 50256, 50256, 50256,
        50256, 50256, 50256, 50256, 50256]), tensor([1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
        0, 0, 0, 0, 0, 0, 0]))
```

Figure 6: GPT2 Tokenizer in action

Here we see that the input prompt containing ciphertext and result has their corresponding tokens. Since we have used a tokenizer of fixed size, the unused indexes have higher integer values and have been masked by an attention mask containing 0 at that index.

The training and validation data was fit into the model each having a batch size of 32 and was run for 5 epochs.

```
Num examples = 27260
Num Epochs = 5
Instantaneous batch size per device = 32
Total train batch size (w. parallel, distributed & accumulation) = 32
Gradient Accumulation steps = 1
Total optimization steps = 4260
Number of trainable parameters = 125198592
```

Figure 7: Training Parameters for the GPT-Neo Fine Tuning

[4260/4260 37:12, Epoch 5/5]

Step	Training Loss	Validation Loss
500	1.920900	1.839412
1000	1.826400	1.815576
1500	1.794200	1.798747
2000	1.760900	1.792532
2500	1.746500	1.784798
3000	1.721100	1.785910
3500	1.708000	1.788620
4000	1.688200	1.787953

Figure 8: Training and Validation Loss obtained every 500 steps

After fine-tuning, we tested the model on our test dataset containing 3407 rows. The model has shown an **overall accuracy of 85.236%**.

```
print(f"Correct: {correct}, Total: {total}")
print(f"Accuracy: {correct/total}")
```

```
Total test dataset size = 3407

Accuracy 0.8324038743762842

Completed 100.0%

Computed 3407 results
Correct: 2836, Total: 3407
Accuracy: 0.8324038743762842
```

Figure 9: Testing Accuracy for the Fine-Tuned GPT-Neo Model

6 FUTURE SCOPE

We would also like to alter the text to matrix conversion for the dataset to make CNN model perform better as it is not performing up to the standards that we want.

We would like to finetune our models to see if we can increase the accuracy. We are also curious to see if we can use an RNN instead of transformers and record the difference in accuracy of the two.

A similar analysis can be performed on other models such as:

- OPT
- BERT
- GPTJ
- Bloom

Apart from this, we would like to explore feature detection to create a model and compare this approach with the transformer-based model.

Once we are satisfied with the model accuracy on 5 classes of ciphers, we would like to extend the model to detect more classes such as AES, DES, RSA, etc.

7 CONCLUSION

This report shows the current progress of our work in the detection of ciphertext using Artificial Neural Networks. Here we used two types of Neural Networks to achieve our objective, namely: Convolution Neural Networks and Transformer approach.

With the CNN approach, our model was able to achieve an accuracy of 61.90% for 5 class classifications. We saw a significant improvement with the transformers approach on the same dataset and classes giving an accuracy of 83.24%. This was primarily due to the higher number of parameters of Gpt-neo and the ability to memorize the local importance of tokens due to the self-attention feature.

In future work, we would like to extend the number of classes of ciphertext and also increase our dataset to find how our models perform in the new environment. We would like to try a feature-learning approach and also use more transformer models. We would also like to solve the Cipher ID challenge from Mystery Twister C3 consisting of 500 ciphertexts. Based on our results and improvements, we would like to publish our work in a journal to see how our results stand in global competition.

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